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To the Editors

Journal of Chemical Information and Modeling

Dear Editors,

We submit for consideration as an **Article** the manuscript “**Ground Truth That Does Not Ground: Substituting Reference Solvent σ -Profiles Degrades COSMO-SAC Solubility Prediction**” by N. L. Polomoshnov and A. V. Rudik.

Physics-informed models are built so that a prediction passes through a quantity chemistry already tabulates, and the design carries an assumption that is rarely tested: that handing the model the tabulated value, in place of the one it learned, should improve it. We test that assumption directly. A graph neural network predicts the σ -profiles a fixed COSMO-SAC layer consumes, and quantum chemistry tabulates the same profiles independently, so the substitution can be made without retraining anything.

It makes the model worse. Over five seeds the mean absolute error on solid–liquid solubility rises from 1.93 to 2.34 in $\ln x_2$, about 21%, with no overlap between the per-seed values of the two arms, and the mean per-solute Spearman correlation over solvents falls by 0.185, a loss in the ranking that solvent selection actually uses. Supervising the same network against the same database during training, the other operation the word “grounding” covers, gives gains that straddle zero, and we report it as having no established sign rather than as a benefit. The operation that fails is the one available in practice, because it is the one that needs no retraining.

The constructive half of the paper is a way to tell in advance where this will happen. Using infinite-dilution activity coefficients we separate a deployed thermodynamic model’s own error from what its inputs leave unresolved, search fifty-nine chemical strata, and find the model’s own error dominant on one solvent chemistry. We state plainly which of those bounds are established and which are the instrument’s output on a set too small to certify, including one that an independent set reproduces at half its size without meeting our own stability rule.

We think this belongs in JCIM for two reasons. It concerns how a widely used cheminformatics construction behaves when a learned component meets a fixed physical map, which is a modelling question rather than a chemistry-specific one; and it is written to the standard of the Journal’s editorial on method and data sharing (*J. Chem. Inf. Model.* **2015**, 55, 719). Every number in the manuscript and the Supporting Information is produced by a deposited script from a deposited artifact; the code, checkpoints, processed splits and per-arm predictions are released; and the two decomposition sets are deposited as row-level machine-readable tables so a reader can recompute the central claim without running our code. Twelve defects and limits of the work are disclosed by us, each where its numbers print, and none of them is reported by anyone else.

This manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have read and approved the final version of the manuscript. We declare no conflicts of interest. This study was supported by the Russian Science Foundation, project No. 25-25-00148.

We would be glad to answer any questions.

Sincerely,
Nikita L. Polomoshnov, on behalf of both authors

Suggested reviewers

- **Fabian Jirasek**, Laboratory of Engineering Thermodynamics, RPTU Kaiserslautern-Landau, Germany. fabian.jirasek@rptu.de
Machine-learning models for activity coefficients under thermodynamic constraints.
- **André Bardow**, Energy and Process Systems Engineering, ETH Zürich, Switzerland. abardow@ethz.ch
Learned predictors for limiting activity coefficients, and physics-guided models constrained to thermodynamic consistency.
- **Ian H. Bell**, Applied Chemicals and Materials Division, NIST, Boulder, Colorado, USA. ian.bell@nist.gov
Author of the open-source reference COSMO-SAC implementation this work validates against.
- **Chieh-Ming Hsieh**, Department of Chemical and Materials Engineering, National Central University, Taoyuan City, Taiwan. hsiehcm@ncu.edu.tw
Author of COSMO-SAC-2010 and of the dispersion term the manuscript implements.
- **William H. Green**, Department of Chemical Engineering, Massachusetts Institute of Technology, USA. whgreen@mit.edu
SolProp and FastSolv, the external points of comparison used here.
- **Hans Hasse**, Laboratory of Engineering Thermodynamics, RPTU Kaiserslautern-Landau, Germany. hans.hasse@rptu.de
Engineering thermodynamics of mixtures, and hard-constrained models for activity coefficients.
- **Kai Sundmacher**, Otto-von-Guericke University Magdeburg and Max Planck Institute for Dynamics of Complex Technical Systems, Germany. sundmacher@mpi-magdeburg.mpg.de
Graph networks for the temperature dependence of activity coefficients.